Velez Control Systems

VCS_SYS1

High-Current Programmable MOSFET Control Circuit + Wide Input 3.3V Output Buck Converter + Raspberry Pi 5 Header Addition

Abstract

VCS_SYS1 is a 6-layer mixed-signal control board designed by Velez Control Systems for high-current programmable load driving with real-time audio-reactive capability. The board is built around a 32-bit ARM Cortex-M0+ microcontroller executing a complete on-chip DSP pipeline: PDM microphone capture via SPI+DMA, third-order CIC decimation, three-band biquad IIR bandpass filtering in fixed-point arithmetic, per-band envelope following, and four-channel PWM output. Each PWM channel drives an N-channel MOSFET through a dedicated half of a dual low-side gate driver, enabling low-impedance switching of high-current loads across four independent channels. The output stage is load-agnostic and suitable for driving LEDs, solenoids, motors, heating elements, or any resistive or inductive DC load within the board's voltage and current ratings.

Power regulation is handled by a wide-input synchronous buck converter, accepting 3.5–28V and providing a stable 3.3V rail for the MCU and peripheral ICs. A discrete MOSFET-based shunt regulator supplies the gate driver VDD from the main power rail. USB connectivity is provided through a USB-UART bridge, supporting firmware programming via the MCU's bootstrap loader. Audio input is captured by a PDM microphone mounted on-board.

The board is designed in a Raspberry Pi 5 HAT form factor with a standard header interface, allowing it to operate as a standalone controller or as a peripheral to a Raspberry Pi 5 host running higher-level processing such as edge AI inference. High-current output is routed through a rated through-hole connector suitable for the expected load.

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1. General Description

The VCS_SYS1 is a compact four-channel control board designed for programmable DC load switching with optional audio-reactive operation. The board accepts a wide DC input range and regulates its own logic supply on-board. Each of the four output channels consists of a gate driver and N-channel MOSFET pair, forming a load-agnostic output stage rated for high continuous current.

In standalone mode, the on-board microcontroller runs a complete audio-to-PWM DSP pipeline, mapping frequency band energy to output channels in real time with no external processing required. When paired with a Raspberry Pi 5 host via the 40-pin HAT header, the board can be commanded externally for custom automation, wireless control, or AI-driven applications.

Firmware is loaded through a USB-UART bridge connected to the MCU's bootstrap loader, requiring no external debug hardware.

2. Features

- Four independent high-current MOSFET output channels
- Load-agnostic output stage: LEDs, motors, solenoids, resistive or inductive DC loads
- Wide input voltage range: 3.5V to 28V
- On-board 3.3V synchronous buck converter
- Discrete shunt regulator for gate driver supply
- Dual low-side gate drivers with low-impedance output
- 32-bit ARM Cortex-M0+ microcontroller
- On-board PDM microphone with full DSP pipeline in firmware
- Three-band frequency decomposition with per-band envelope following
- Four-channel PWM generation with adjustable frequency and duty cycle
- USB-UART bridge for firmware programming via bootstrap loader
- Serial interface for runtime diagnostics and configuration
- Raspberry Pi 5 HAT form factor with 40-pin header
- Standalone or host-controlled operation
- 6-layer PCB construction
- Rated through-hole connector for high-current output
- Single-supply operation, no external regulators required

3. Applications

- Audio-reactive LED and lighting installations
- Architectural and stage lighting control
- RGB and RGBW LED strip driving
- DC motor speed control
- Solenoid and valve actuation
- Resistive heating element control
- Automated test fixtures requiring programmable DC switching
- Robotics and mechatronics output stages
- IoT and edge AI actuator platforms
- Raspberry Pi 5 peripheral expansion for high-current loads
- Prototyping and development of custom embedded control systems

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4. Absolute Maximum Ratings

Stresses beyond those listed may cause permanent damage to the board. These are stress ratings only; operation at or near absolute maximum conditions is not recommended.

Parameter	Symbol	Min	Max	Unit
Board input voltage	V_IN	−0.3	28	V
MCU supply voltage (VDD pin)	V_DD	−0.3	4.1	V
MCU I/O voltage (standard pins)	V_IO	−0.3	V_DD + 0.3	V
Gate driver supply voltage	V_DRV	−0.3	18	V
MOSFET drain-source voltage	V_DS	—	40	V
MOSFET continuous drain current (per ch, T_C = 25°C)	I_D	—	50	A
MOSFET pulsed drain current	I_DM	—	200	A
MOSFET gate-source voltage	V_GS	−20	+20	V
USB VBUS voltage	V_USB	—	V_IO + 2.5	V
Storage temperature	T_STG	−55	175	°C
MCU junction temperature	T_J	−40	150	°C

5. Recommended Operating Conditions

Parameter	Symbol	Min	Typ	Max	Unit
Input supply voltage	V_IN	3.5	12	28	V
Regulated logic rail	V_3V3	3.2	3.3	3.4	V
MCU operating voltage	V_DD	1.62	3.3	3.6	V
Gate driver supply voltage	V_DRV	4.5	—	18	V
Buck converter output current	I_BUCK	—	—	3	A
MOSFET continuous drain current	I_D	—	—	50	A
Ambient operating temperature	T_A	−40	25	125	°C
PDM microphone supply voltage	V_MIC	1.65	3.3	3.63	V
USB-UART bridge supply voltage	V_UART	3.0	3.3	3.6	V

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6. Electrical Characteristics

6.1 Power Supply

Parameter	Condition	Min	Typ	Max	Unit
Input voltage range	—	3.5	—	28	V
Board quiescent current	V_IN=12V, no load	—	TBD	—	mA

6.2 Buck Converter

Parameter	Condition	Min	Typ	Max	Unit
Input voltage	—	3.5	—	28	V
Output voltage	—	—	3.3	—	V
Output current	Continuous	—	—	3	A
Switching frequency	—	—	570	—	kHz
Efficiency	V_IN=12V, I_OUT=1A	—	~90	—	%
Output voltage ripple	V_IN=12V, I_OUT=3A	—	TBD	—	mV
Inductor (SRN6045-470M)	—	—	47	—	µH
Inductor DCR	—	—	257	—	mΩ
Inductor saturation current	—	—	—	1.3	A
Inductor rated current	—	—	—	1.1	A
Catch diode fwd voltage	1N5819, I_F=1A	—	—	0.6	V
Catch diode reverse voltage	—	—	—	40	V

6.3 MOSFET Output Stage

Parameter	Condition	Min	Typ	Max	Unit
Drain-source voltage	—	—	—	40	V
Continuous drain current	V_GS=10V, T_C=25°C	—	—	50	A
On-resistance	V_GS=10V, I_D=20A	—	—	7	mΩ
On-resistance	V_GS=4.5V, I_D=15A	—	—	9.5	mΩ
Total gate charge	V_GS=10V, V_DS=20V	—	—	80	nC
Number of output channels	—	—	4	—	—
Package	—	—	TO-252 (DPAK)	—	—

6.4 Gate Drivers

Parameter	Condition	Min	Typ	Max	Unit
Supply voltage	V_DD	4.5	—	18	V
Peak source current	—	—	5	—	A
Peak sink current	—	—	5	—	A
Propagation delay	—	—	17	—	ns
Channel delay matching	—	—	1	—	ns

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Parameter	Condition	Min	Typ	Max	Unit
Configuration	—	—	Dual non-inv, low-side	—	—
UVLO threshold	—	—	5	—	V
Operating temperature	—	−40	—	140	°C
Number of ICs	—	—	2	—	—

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6.5 Microcontroller

Parameter	Condition	Min	Typ	Max	Unit
Core	—	—	ARM Cortex-M0+	—	—
Maximum clock frequency	—	—	80	—	MHz
Flash memory	—	—	128	—	KB
SRAM	—	—	32	—	KB
Supply voltage	—	1.62	3.3	3.6	V
ADC resolution	—	—	12	—	bits
ADC sample rate	—	—	4	—	Msps
Number of ADC channels	External	—	17	—	—
DAC resolution	—	—	12	—	bits
Operating temperature	—	−40	—	125	°C
Package	—	—	QFP-48	—	—

6.6 USB-UART Bridge

Parameter	Condition	Min	Typ	Max	Unit
USB standard	—	—	USB 2.0 Full Speed	—	—
Maximum baud rate	—	—	1.5	—	Mbps
Supply voltage	—	3.0	3.3	3.6	V
Supply current	—	—	9.5	—	mA
Operating temperature	—	−40	—	85	°C
Package	—	—	QFN-20 (3×3 mm)	—	—

6.7 PDM Microphone

Parameter	Condition	Min	Typ	Max	Unit
Type	—	—	MEMS, digital PDM	—	—
Supply voltage	—	1.65	3.3	3.63	V
Sensitivity	1 kHz, 94 dB SPL	—	−32	—	dBFS
Sensitivity tolerance	—	—	±1	—	dB
SNR	—	—	64	—	dB
Directivity	—	—	Omnidirectional	—	—
Port location	—	—	Bottom	—	—
Package	—	—	3.50 × 2.65 mm	—	—

6.8 Shunt Regulator

Parameter	Condition	Min	Typ	Max	Unit
Zener voltage	I_ZT = 10 mA	—	17	—	V

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Parameter	Condition	Min	Typ	Max	Unit
Zener tolerance	—	—	±5	—	%
Zener power dissipation	—	—	—	500	mW
Zener package	—	—	SOD-123	—	—
Shunt MOSFET	—	—	AOD4184A	—	—

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7. Block Diagram

Refer to the inline block diagram provided separately. A high-resolution version will be included in a future revision of this document.

8. Pin Configuration and Functions

8.1 Raspberry Pi 5 HAT Header

The board includes a 40-pin Samtec SSW-120-01-G-D-LL dual-row header matching the Raspberry Pi 5 HAT standard. The buck converter 3.3V output is permanently connected to header pins 1 and 17, providing regulated power to an attached Pi at all times. The 5V pins (2, 4) are passed through from the Pi's own supply.

Jumper headers J1, J9, J10, and J11 allow the user to connect or disconnect PWM and GPIO lines between the Pi header and the gate driver inputs. With jumpers installed, the Raspberry Pi can directly drive the MOSFET output channels, bypassing the on-board MCU. With jumpers removed, the gate drivers are controlled exclusively by the MCU's PWM outputs.

Pin	Name	Pin	Name
1	3.3V (Buck Output)	**2**	5V
3	SDA	**4**	5V
5	SCL	**6**	GND
7	GPCLKD	**8**	TXD
9	GND	**10**	RXD
11	GPIO17	**12**	PCM_CLK
13	GPIO27	**14**	GND
15	GPIO22	**16**	GPIO_23
17	3.3V (Buck Output)	**18**	GPIO_24
19	MOSI	**20**	GND
21	MISO	**22**	GPIO_25
23	SCLK	**24**	CE0
25	GND	**26**	CE1
27	ID_SD	**28**	ID_SC
29	GPIO5	**30**	GND
31	GPIO6	**32**	PWM0
33	PWM1	**34**	GND
35	PCM_FS	**36**	GPIO16
37	GPIO26	**38**	PCM_DIN
39	GND	**40**	PCM_DOUT

Jumper Headers

Jumper	Function
J10	Connects/disconnects Pi GPIO to gate driver input channel(s)
J11	Connects/disconnects Pi GPIO to gate driver input channel(s)
J9	Connects/disconnects Pi PWM/GPIO to gate driver input channel(s)
J1	Connects/disconnects Pi PWM/GPIO to gate driver input channel(s)

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Note: With all jumpers removed, the gate drivers receive PWM only from the MCU. With jumpers installed, the Pi can override MCU control of the corresponding output channels. Do not drive the same gate driver input from both the MCU and the Pi simultaneously.

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8.2 Load Output Connector

The high-current load output uses a JST VH series 4-pin through-hole connector. Each pin corresponds to one MOSFET-switched output channel. All channels share a common return path through the board ground plane.

Pin	Channel	MOSFET	Function
1	R	Q1	Channel 1 output (red / load 1)
2	G	Q2	Channel 2 output (green / load 2)
3	B	Q3	Channel 3 output (blue / load 3)
4	W	Q4	Channel 4 output (white / load 4)

8.3 Power Input Connector

The DC barrel jack (PJ-002AH-SMT-TR) accepts a wide-range DC input supplying the entire board. Center pin is positive. Voltage range: 3.5V to 28V DC. Supplies the buck converter input, shunt regulator input, and MOSFET drain rail.

8.4 USB Connector

The USB-C connector (TYPE-C-31-M-12) provides the host interface for firmware programming and serial communication through the CP2102N USB-UART bridge. USB 2.0 Full Speed. VBUS is not used for board power; used for USB-UART bridge supply detection only.

8.5 Tactile Switch

A single on-board tactile switch (TS04-66-50-BK-160-SMT) is provided for user-defined input. Momentary, normally open. Default assignment: BSL invoke / user-configurable.

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9. Detailed Description

9.1 Power Architecture

The VCS_SYS1 operates from a single DC input ranging from 3.5V to 28V, supplied through an SMD barrel jack connector. This input rail feeds two independent regulation paths: a synchronous buck converter producing a regulated 3.3V rail for all digital and low-voltage analog circuitry, and a discrete shunt regulator providing a separate supply for the gate driver ICs.

The buck converter uses an externally compensated current-mode topology switching at 570 kHz. The output filter consists of a 47 µH semi-shielded wirewound inductor and ceramic output capacitors. A Schottky catch diode provides the freewheeling path. The 3.3V rail supplies the MCU, USB-UART bridge, PDM microphone, and logic-level pull-ups. The converter is rated for up to 3A continuous output, though typical board consumption is well below this under normal operating conditions.

The input rail also directly supplies the drain side of all four output MOSFETs. The load current path is therefore independent of the regulated 3.3V rail and is limited only by the input supply capability, the MOSFET ratings, and the PCB copper capacity.

9.2 Gate Driver and MOSFET Output Stage

Each output channel consists of one half of a dual low-side gate driver IC driving a single N-channel MOSFET. Two dual gate driver ICs provide four total drive channels. The gate drivers accept 3.3V logic-level PWM inputs from the MCU and translate them to the full gate drive voltage set by the shunt regulator, ensuring the MOSFETs are driven into full enhancement regardless of the logic rail voltage.

The gate drivers are capable of sourcing and sinking 5A peak current into the MOSFET gate capacitance, resulting in fast switching transitions and low switching losses. Propagation delay is typically 17 ns with 1 ns channel-to-channel matching, making the four channels suitable for synchronized or phase-shifted drive applications.

Each MOSFET is an N-channel device in a TO-252 (DPAK) package rated for 40V drain-source voltage and 50A continuous drain current at 25°C case temperature. On-resistance is less than 7 mΩ at 10V gate-source voltage. The total gate charge is approximately 80 nC per device. Five MOSFETs are populated on-board: four serving the output channels and one used in the shunt regulator circuit.

9.3 Shunt Regulator

The gate driver VDD supply is derived from the main DC input rail through a discrete MOSFET-based shunt regulator. A 17V Zener diode sets the clamp voltage, and a series resistor limits current into the shunt element. One of the AOD4184A MOSFETs is configured as the shunt pass device. This architecture ensures the gate drivers receive a supply voltage high enough for full MOSFET enhancement across the full input voltage range, while dissipating excess voltage as heat in the shunt element rather than requiring a dedicated high-voltage linear regulator.

The shunt regulator output is decoupled with local bypass capacitors at each gate driver VDD pin. The Zener diode is rated for 500 mW and is packaged in a SOD-123 footprint.

9.4 Audio Input Path

Audio input is provided by an on-board MEMS PDM microphone. The microphone is an omnidirectional, bottom-port device with a digital pulse-density modulated output. Sensitivity is −32 dBFS at 94 dB SPL with ±1 dB tolerance, and the signal-to-noise ratio is 64 dB. The microphone operates from the 3.3V rail and requires only a clock input and a single data output line, both routed directly to the MCU.

The PDM clock is generated by the MCU’s SPI peripheral operating in master mode, clocking at 2.4 MHz. The PDM data line is captured on the SPI MISO input using DMA, providing continuous audio acquisition with minimal CPU overhead.

9.5 DSP Pipeline

All audio processing is performed on-chip by the MCU in firmware. The raw PDM bitstream captured via SPI+DMA is first decimated by a third-order cascaded integrator-comb filter with a decimation ratio of 64, converting the 2.4 MHz single-bit stream into a multi-bit PCM sample stream at approximately 37.5 kHz. The decimated samples are then split into three parallel paths, each containing a second-order biquad IIR bandpass filter targeting a distinct frequency band: bass, midrange, and high frequencies. All filter arithmetic is performed in Q15 or Q14 fixed-point to avoid floating-point overhead on the Cortex-M0+ core.

Each bandpass output feeds an envelope follower that tracks the instantaneous energy in its frequency band, producing a slowly varying control signal. The three envelope outputs are mapped to PWM duty cycles on three of the four output channels. The fourth channel can be assigned to a composite or user-defined function. PWM generation uses the MCU’s on-chip timer peripherals, with adjustable frequency and resolution.

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9.6 MCU Peripherals and Configuration

The MCU is an 80 MHz ARM Cortex-M0+ device with 128 KB flash and 32 KB SRAM. SPI0 is configured as master for PDM microphone clock generation and data capture, operating at 2.4 MHz with DMA on the receive channel. Four timer channels generate independent PWM outputs routed to the gate driver inputs. UART0 is connected to the CP2102N USB-UART bridge for BSL firmware programming and runtime serial communication. Additional GPIO pins are allocated to the tactile switch input and any user-configurable jumper-routed signals.

The MCU operates from the regulated 3.3V rail with local bypass capacitance on VDD and VCORE pins. The internal SYSOSC provides the 80 MHz clock source, requiring no external crystal for standard operation.

9.7 USB-UART and BSL Programming

The CP2102N USB-UART bridge provides a USB 2.0 Full Speed interface to the host PC, translating USB transactions to UART at baud rates up to 1.5 Mbps. The bridge is connected to the MCU's UART0 peripheral and supports the MSPM0G3507's bootstrap loader protocol, allowing firmware download without an external debug probe or JTAG connection.

The USB-C connector carries the USB data lines to the CP2102N. VBUS is used only for bridge supply detection and does not power the board. A 1 kΩ pull-up on the bridge reset line is tied to the 3.3V rail. The on-board tactile switch can be used to invoke BSL mode by holding it during power-on or reset.

9.8 Raspberry Pi 5 Host Interface

The board conforms to the Raspberry Pi HAT mechanical specification and includes a 40-pin dual-row header matching the standard Pi GPIO pinout. The regulated 3.3V buck output is permanently connected to the header's 3.3V pins (pins 1 and 17), providing power to an attached Raspberry Pi 5 at all times when the board is energized.

The MCU and the Pi do not share any direct communication bus. Instead, jumper headers J1, J9, J10, and J11 provide selectable signal paths between specific Pi GPIO pins and the gate driver inputs. When these jumpers are installed, the Pi can directly drive the MOSFET output stages using its own PWM or GPIO outputs, effectively replacing the MCU as the control source for some or all output channels. When the jumpers are removed, the gate drivers are controlled exclusively by the MCU.

This architecture supports three operating configurations: standalone MCU control with the Pi serving only as a powered compute platform (jumpers removed), full Pi-driven output control (all jumpers installed, MCU PWM disabled or tristated), or a hybrid configuration where the Pi controls a subset of channels and the MCU controls the rest (selected jumpers installed). In all cases, only one source should drive a given gate driver input at a time to avoid bus contention.

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10. Application Information

10.1 Standalone Audio-Reactive Operation

In its default configuration, the VCS_SYS1 operates as a fully self-contained audio-reactive controller. The on-board PDM microphone captures ambient sound, and the MCU's DSP pipeline decomposes the audio into three frequency bands in real time. Each band's energy envelope is mapped to a PWM duty cycle on a corresponding output channel, producing a dynamic, music-responsive drive signal without any external processing or host connection.

To use the board in this mode, connect a DC supply within the rated input range to the barrel jack, connect the load to the JST VH output connector, and remove all Pi-to-gate-driver jumpers (J1, J9, J10, J11). The MCU will begin processing audio and driving outputs immediately on power-up, assuming firmware has been previously loaded via the USB-C port.

The frequency band boundaries, envelope follower time constants, and PWM-to-duty-cycle mapping are all configurable in firmware. The fourth output channel is available for a user-defined function such as a broadband composite, a static bias level, or an externally commanded override via UART.

10.2 Host-Controlled Operation via Raspberry Pi 5

When a Raspberry Pi 5 is mounted to the HAT header, the board provides regulated 3.3V power to the Pi at all times. By installing the appropriate jumpers (J1, J9, J10, J11), the Pi's PWM-capable GPIO lines are routed directly to the gate driver inputs, allowing the Pi to control some or all of the four output channels.

In this configuration, the MCU's PWM outputs to the affected channels should be tristated or disabled in firmware to avoid bus contention. The Pi can then drive the outputs using standard GPIO libraries at whatever update rate and logic its application requires. Typical use cases include edge AI wake word detection triggering lighting scenes, wireless control via the Pi's networking stack, scheduled automation routines, or closed-loop feedback systems using Pi-connected sensors.

In a hybrid configuration, the MCU retains control of a subset of channels for low-latency audio-reactive response while the Pi manages the remaining channels for higher-level behavior.

10.3 Load Selection and Wiring Guidelines

The output stage is load-agnostic. Any DC load that operates within the board's voltage and current ratings may be connected. For resistive loads such as LED strips and heating elements, no special precautions are required beyond verifying that the load's operating voltage and current fall within the MOSFET and connector ratings. The load connects between the positive supply rail and the MOSFET drain output on the JST VH connector. The MOSFET switches the low side to ground.

For inductive loads such as solenoids, relays, and DC motors, a freewheeling diode must be installed across the load terminals as close to the load as possible. The diode cathode connects to the positive supply and the anode to the MOSFET drain. This clamps the inductive kick at turn-off and protects the MOSFET from voltage spikes exceeding the 40V drain-source rating. A fast-recovery or Schottky diode rated for the load current is recommended.

Wire gauge between the board and the load should be selected for the expected continuous current. For runs longer than 30 cm at currents above 5A, use a minimum of 16 AWG stranded wire. Keep the return path (ground) wire gauge equal to or larger than the load wire to minimize voltage drop and ground bounce.

10.4 Thermal Considerations

The primary heat-generating components on the board are the four output MOSFETs and the buck converter. Under typical operating conditions with moderate load currents, no additional heatsinking is required.

Each output MOSFET dissipates power according to $P = I^2 \cdot R_{DS(on)}$, where $R_{DS(on)}$ is approximately 7 mΩ at full gate enhancement. At 10A continuous per channel, this yields approximately 0.7W per

MOSFET. The DPAK package provides a thermal path through the drain pad to the PCB copper, and the 6-layer board construction offers substantial copper area for heat spreading.

The shunt regulator dissipates power proportional to the difference between the input voltage and the 17V Zener clamp voltage, multiplied by the gate driver quiescent current draw. At high input voltages (24–28V) this dissipation increases and should be accounted for in the thermal budget. For enclosed installations or sustained high-current operation, ensure adequate ventilation around the MOSFET and converter areas. Forced air cooling is recommended for applications exceeding 20A total continuous load current.

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11. PCB Layout

Reserved for future revision.

12. Mechanical Dimensions

Reserved for future revision. Board conforms to Raspberry Pi HAT mechanical specification.

13. Bill of Materials

Ref Des	Description	Package	Qty	LCSC Part
J5	DC barrel jack	SMD	1	C22434687
U2	Buck converter, 3.5–28V in, 3A	SOIC-8	1	C2872693
U3, U1	Dual low-side gate driver, 5A/5A	SOIC-8	2	C465729
U4	ARM Cortex-M0+ MCU, 80 MHz, 128KB	QFP-48	1	C22389960
U5	USB-UART bridge, USB 2.0	QFN-20	1	C969913
U6	Zener diode, 17V, 500 mW	SOD-123	1	C236110
Q1–Q4	N-channel MOSFET, 40V, 50A	DPAK	4	C99124
Q5	N-channel MOSFET (shunt reg)	DPAK	1	C99124
MIC	PDM MEMS microphone, −32 dBFS	3.50×2.65 mm	1	C3171752
USBC	USB-C connector	SMD	1	C165948
S1	Tactile switch, momentary	SMD	1	C4364861
L1	Inductor, 47 µH, 1.1A	6.0×6.0 mm	1	C2041480
D1	Schottky diode, 40V, 1A	DO-41	1	C437199
R5	47 kΩ	0402	1	C413103
R6	1 kΩ	0603	1	C14676
R7	22 Ω	1206	1	C114943
R8, R9	31.6 kΩ	0603	2	C185339
R11	47.5 kΩ	0402	1	C137974
R12, R13	5.1 kΩ	0603	2	C105580
R15	200 Ω	2512	1	C326622
C1, C5	100 nF	0805	2	C16195247
C2	470 nF	0402	1	C1623
C3	1 µF	0805	1	C1712
C4	18 nF	0805	1	C162313
C6	4.7 µF	0402	1	C1705
C7,C15,C16,C17	100 nF	0603	4	C77055
C8	10 nF	0402	1	C1589
C9	68 pF	0402	1	C28262
C10	100 nF	0402	1	C1591
C11, C14	10 µF	0805	2	C15850
C12	4.7 nF	0402	1	C1621
C13	4.7 µF tantalum	SMD	1	C60867
C18	1 µF	0402	1	C15849

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14. Revision History

Revision	Date	Description
1.0	March 2026	Initial release

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